Understanding the Karno tank signals: a statistical route

following Madsen, *Time Series Analysis*

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1 Context

The grey-box model (`scripts/graybox.R`, `ctsmTMB`) treats the storage tank as two lumped thermal masses, T_{top} and T_{bot}, coupled by a linear conductive term $k\,(T_{\text{top}} - T_{\text{bot}})$, driven by the air-source heat pump's electrical power P_{el} (into the top node) and the district-heating draw $\dot{Q}_{\text{dist}}$ (at the bottom node), with a loss term $UA\,(T - T_{\text{amb}})$ on each node.

Fitting this model to ∼11 days of real logged data (`data/karno-410708_raw_k0001.parquet`, 2026-05-13 to 2026-05-24, 5-minute sampling) produces a badly behaved fit: several parameters (C_{top}, C_{bot}, k, a) pin at whatever bound they are given rather than converging to an interior optimum, and the measurement/process noise parameters split asymmetrically between the two channels ($\sigma_{y_1} \to$ its upper bound, $\sigma_{y_2} \to$ its lower bound, or vice versa depending on how the bounds are set) — the classic sign of the optimizer using one channel's noise term to paper over a structural mismatch instead of a real fit.

This document is not a fix. It is a statistical route — grounded in Madsen's *Time Series Analysis* — for actually characterizing the input/output dynamics of the two sensors *before* committing to any particular ODE structure again.

1.1 Physical layout

top of tank
| 30 cm
T_{top} sensor
| 70 cm
T_{bot} sensor
| 30 cm
bottom of tank

Heat is added at the T_{top} level; the district-heating draw ("cool") is taken out at the T_{bot} level. Despite this asymmetric but plausible geometry, the observed behaviour is not what a simple stratified/conductive model predicts:

- T_{top}: flat, then declines with no heat-pump input; rises when the heat pump runs. This is expected – it is the injection point.
- T_{bot}: rises *directly* when the heat pump runs, with no visible delay consistent with a thermocline having to migrate 70 cm down past the sensor.

2 The route

The steps below are ordered the way they should actually be executed: each step's output is required input for the next. Section numbers refer to Madsen, *Time Series Analysis*.

Step 0 — univariate structure

[Madsen, §5.5, 6.2.1]

Before touching cross-relationships, look at each series alone: autocorrelation function (ACF) and partial ACF (PACF) of T_{top}, T_{bot}, P_{el}, $\dot{Q}_{\text{dist}}$ individually (§6.2.1, estimation of the ACF; §5.5 for what AR/MA signatures look like in the ACF/PACF). At 5-minute sampling all four series will be highly persistent (ACF decaying slowly, near unit-root behaviour). This matters directly for Step 2: it is exactly what makes a *raw* cross-correlation between two persistent series unreliable.

Step 1 — raw cross-correlation function (CCF)

[Madsen, §6.2.2]

The cross-covariance and cross-correlation functions,

$$\hat{\gamma}_{xy}(\ell) = \frac{1}{n}\sum_t (x_t - \bar{x})(y_{t+\ell} - \bar{y}), \qquad \hat{\rho}_{xy}(\ell) = \frac{\hat{\gamma}_{xy}(\ell)}{\sqrt{\hat{\gamma}_{xx}(0)\,\hat{\gamma}_{yy}(0)}},$$

computed for a range of lags ℓ, is the formal version of "lag correlation". Do this first as a baseline, but treat the result with suspicion: two autocorrelated series will show spuriously significant CCF values at many lags purely because of their own internal persistence, not because one drives the other. This is why Step 2 exists.

Step 2 — pre-whitening

[Madsen, §9.6.1]

The standard (Box–Jenkins) fix for the Step 1 problem:

(i) Fit a model (AR(p) is enough in practice) to the input series x_t such that the residuals $a_t = x_t - \sum_i \phi_i x_{t-i}$ are approximately white noise.

(ii) Apply the *same* filter to the output series: $b_t = y_t - \sum_i \phi_i y_{t-i}$.

(iii) Compute the CCF between a_t and b_t.

Because a_t is white, the resulting cross-correlation is a clean, interpretable estimate of the dynamic relationship between x and y – its shape approximates the system's impulse response (Step 3), and its confidence bands are the simple $\pm 1.96/\sqrt{n}$ used for white-noise input.

This has already been run (`scripts/diagnostics.R`) for $P_{\text{el}} \to T_{\text{top}}$ and $P_{\text{el}} \to T_{\text{bot}}$. Results, with the convention that a positive lag means the input leads the output:

Pair	AR order	peak lag	peak r
$P_{\text{el}} \to T_{\text{bot}}$	20	+1 sample (+5 min)	+0.296
$P_{\text{el}} \to T_{\text{top}}$	20	0 samples (0 min)	−0.393

$P_{\text{el}} \to T_{\text{bot}}$ is exactly what a direct heat input should look like: small positive lag, positive sign. $P_{\text{el}} \to T_{\text{top}}$ is not: the strongest relationship is *instantaneous and negative*, at no lag where a delayed positive response would be expected from direct heat injection at that sensor's level.

Step 3 — reading the CCF as an impulse response

[Madsen, §8.3–8.4]

Once pre-whitened, the shape of the cross-correlation function *is* a (scaled) estimate of the system's impulse-response weights $v_0, v_1, v_2, \ldots$ relating input to output (§8.3.1, transfer function models; §8.4, identification of transfer function models from the CCF). In practice: a single sharp spike at some lag ℓ_0 suggests a simple gain plus dead time of ℓ_0 steps; a spike that decays geometrically over several lags suggests a first-order lag (single time constant); an oscillating or sign-changing pattern (which is what both the raw data and the model residuals show here, see Step 5) suggests either an under-differenced/near-unit-root series contaminating the estimate, or – more likely given the context – a genuinely mis-specified relationship that no simple transfer function order will resolve, because the assumed causal direction is wrong (see §3).

Step 4 — multiple-input extension

[Madsen, §8.5]

There are four real candidate inputs (P_{el}, $\delta =$ GSHP on/off, $\dot{Q}_{\text{dist}}$, T_{amb}). §8.5 covers identifying and estimating transfer functions with several regressors simultaneously (moment relations §8.5.1, spectral relations §8.5.2, identification §8.5.3) rather than one input at a time, which matters once individual pairwise CCFs are understood and need to be combined without double-counting shared variance (e.g. P_{el} and δ are correlated with each other, since both indicate heat-pump activity).

Step 5 — residual analysis of the currently fitted model

[Madsen, §6.6]

For the model that is already fitted (`fit` object from `model$estimate()` in `ctsmTMB`), the one-step-ahead standardized innovations (`fit$residuals$normalized`) should behave like white noise *if* the state equations correctly capture the system's dynamics. Two checks (§6.6.2, residual analysis):

(a) ACF of each residual series alone – significant autocorrelation means the state's own dynamics (not just the inputs) are under-specified.

(b) Cross-correlation of each (pre-whitened, Step 2) input against the residual series – significant correlation at some lag means that input's effect is not fully captured by the fitted model at that lag.

Already run. Result: both residual series show strong, alternating-sign, slowly-decaying autocorrelation out past lag 40 (200 minutes) – the current fit is a poor one-step predictor for both states, not just T_{top}. However, cross-correlating the residuals against P_{el} shows *no* significant lag for either channel. That combination is informative: the problem is not a missing lagged copy of P_{el} in the state equation (which residual cross-correlation would have caught) – it is something the current noise-parameter degeneracy (Step-2 finding) is already masking. Re-run this check once the noise-parameter pathology is resolved (i.e. interior, non-bound-pinned σ estimates), since a degenerate Kalman gain distorts residual whiteness independently of any missing input dynamics.

Step 6 — frequency domain cross-check

[Madsen, §7.3]

As a complement to the time-domain CCF: the cross-spectrum (§7.3) decomposes the input/output relationship by frequency. The *phase spectrum* (§7.3.2) gives lag as a function of frequency (useful if the delay itself is frequency-dependent, e.g. short bursts of heat-pump operation propagate differently than long ones), and the *coherence spectrum* gives, per frequency, how much of the output's variance the input actually explains – useful here because δ (GSHP on/off) is a sparse, near-impulsive signal (duty cycle $\approx 0.5\%$, 5 transitions in 11 days) while P_{el} is closer to continuous; the two may show up very differently in a coherence plot.

Step 7 — the worked example in the book

[Madsen, §1.1.3]

Section 1.1.3, "Heat dynamics of a building", is the book's own example of grey-box RC-network modeling of a thermal system – structurally the same model class used here. Worth reading directly for how the book frames identifiability and model-order selection for this kind of problem.

3 A caveat outside the book's scope

Steps 1–6 (and the whole of Chapter 8) implicitly assume the input is *exogenous* – something that happens independently of the state being modeled. The Step-2 result above (instantaneous, negative $P_{\mathrm{el}} \to T_{\mathrm{top}}$ correlation, versus the physically sane delayed positive $P_{\mathrm{el}} \to T_{\mathrm{bot}}$ correlation) is not consistent with an exogenous input at all: it is the signature of *closed-loop* behaviour, where P_{el} is itself a function of T_{top} via the heat pump's thermostat (power ramps up when T_{top} is low, throttles down as it recovers). This was confirmed directly: the heat pump's on/off command `z_HP,air` correlates 0.95 with P_{el} (expected) but *also* -0.48 with T_{top} – the command itself is triggered by low T_{top}, and the setpoint `T_set,HP,air` is constant ($= 0$) in this window, so there is no clean alternative exogenous proxy available here.

Madsen's book, being a classical (largely open-loop) time series text, does not cover closed-loop identification (see e.g. Ljung, *System Identification: Theory for the User*, for direct/indirect closed-loop methods). What its tools *did* do is surface the problem cleanly – the wrong-signed, zero-lag CCF peak is exactly what closed-loop feedback looks like under this diagnostic. Any structural fix (modeling the thermostat explicitly, restricting to genuinely open-loop stretches of the log, or treating T_{top} as a measured boundary condition rather than a state to be solved for) is a separate design decision, deliberately left open here.

4 Practical checklist

1. Run Step 0 (ACF/PACF) on all four series to confirm the persistence that motivates pre-whitening.
2. Extend `scripts/diagnostics.R`'s pre-whitened CCF (already covers P_{el}, $\dot{Q}_{\mathrm{dist}}$) to δ and T_{amb} as well.
3. Read off Step 3 (impulse-response shape) for whichever pairs come back with a clean, physically sane sign and lag – T_{bot}'s channel looks like the promising one.
4. Re-run Step 5 (residual analysis) once the model's noise parameters are no longer bound-pinned, to separate genuine missing dynamics from Kalman-filter noise-mismatch artifacts.
5. Run Step 6 (coherence/phase spectrum) as a second opinion on Step 2, particularly for δ given how sparse it is.

6. Treat P_{el} (and δ/`z_HP,air`) as provisionally endogenous until/unless an open-loop stretch of data or an explicit controller model says otherwise.